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The asymmetric expression of HSPA2 in blastomeres governs the first embryonic cell-fate decision

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eLife Assessment

This **useful** study by Gao et al identifies Hspa2 as a heterogeneous transcript in the early embryo and proposes a plausible mechanism showing interactions with Carm1. The authors propose that variability in HSPA2 levels among blastomeres at the 4-cell stage skews their relative contribution to the embryonic lineage. Given only 4 other heterogeneous transcripts/non-coding RNA have been proposed to act similarly at or before the 4-cell stage, this would be a key addition to our understanding of how the first cell fate decision is made. While this is a **solid** study, further data are needed to fully support the conclusions.

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Abstract

The first cell-fate decision is the process by which cells of an embryo take on distinct lineage identities for the first time, thus representing the beginning of developmental patterning. Here, we demonstrate that the molecular chaperone heat shock protein A2 (HSPA2), a member of the 70 kDa heat shock protein (HSP70) family, is asymmetrically expressed in the late 2-cell stage of mouse embryos. The knockdown of *Hspa2* in one of the 2-cell blastomeres prevented its progeny predominantly toward the inner cell mass (ICM) fate. In contrast, the overexpression of *Hspa2* in one of the two-cell blastomeres did not induce blastomeres to differentiate towards the ICM fate. Furthermore, we demonstrated that HSPA2 interacts with CARM1 and its levels correlate with ICM-associated genes and that it interacts with the CARM1. Collectively, our results identify HSPA2 as a critical early regulator of the first cell-fate decision in mammalian 2-cell embryos.

1. Introduction

The preimplantation development of mammalian embryos is a complex and orderly process[1]. The first cell-fate decision is a key event in the preimplantation development in mammalian embryos. Specifically, the first cell-fate decision occurs at the morula stage following embryonic compaction and results in the division of embryonic cells into two distinct lineages: the trophectoderm (TE) and the inner cell mass (ICM). The TE is a single layer surrounding the fluid-filled cavity known as the blastocoel that provides extraembryonic structures such as the placenta, whereas the ICM, which is attached to the inside of the TE, contains pluripotent cells that give rise to the fetus and extraembryonic tissues [2–4]. However, it remains an open question of how these cells shape the first cell-fate decision of preimplantation development, and when they begin to trigger the initial differentiation signals.

The asymmetry of the blastocyst may be traced back to the asymmetric nuclear abundance of coactivator-associated arginine methyltransferase 1 (CARM1) at the 4-cell stage [5–7]. The heterogeneous activity of CARM1 leads to the differential methylation of H3 methylation of arginine 26 (H3R26me). More extensive methylation of H3R26me can result in a higher increase in the expression levels of ICM-specific genes (e.g., *Oct4*, *Nanog* and *Sox2*) and global chromatin accessibility, which directs their progeny into the ICM fate [8]. Recent studies have further suggested that asymmetry of the blastomere can even be traced back to the 2-cell stage. For example, a long non-coding RNA, *LincGET*, is known to be asymmetrically expressed in the nuclei of blastomeres in 2 and 4-cell embryos. Specifically, *LincGET* and CARM1 are known to form a complex that promotes the H3 methylation of arginine 26 (H3R26me) and activates the expression of genes in the ICM [9]. In addition, HMGA1, a member of the high mobility group (HMG) protein family, is also known to exhibit heterogeneity as early as the two-cell stage and involves in cell fate differentiation [10]. Collectively, these studies suggest that molecular heterogeneity takes place in the early stage of embryonic development and is associated with cell fate differentiation.

Despite these findings, the specific factors that exhibit heterogeneity at the two-cell stage and the mechanisms by which the cell fate and differentiation can be decided still require further investigation. The molecular chaperone heat shock protein A2 (HSPA2) is a member of the 70 kDa heat shock protein (HSP70) family that is evolutionarily conserved in human and metazoan. It is originally identified in male germ cells and described as a testis-specific protein that is essential for spermatogenesis [11, 12]. The *Hspa2* gene is located on chromosome 14 (14q24.1)[13], and the aberrant expression of HSPA2 in testes was shown to induces primary spermatocytes to arrest in meiosis I and undergo apoptosis, thus leading to male infertility [14, 15]. HSPA2 is expressed in somatic tissues and during mouse embryogenesis [16, 17], and high levels of HSPA2 expression play a critical role in the genesis and progression of carcinoma [18, 19]. However, the role of HSPA2 in lineage differentiation of embryonic cells is yet to be unveiled.

In this study, we separated blastomeres from the 4-cell stage and cultured to the morula stage, analyzed the gene expression pattern by using Smart-seq2 to identify differentially expressed genes. *Hspa2* was identified and was shown to be asymmetrically distributed in two-to four-cell blastomeres. The knockdown of *Hspa2* in one of the 2-cell blastomeres prevented its progeny predominantly towards the ICM fate. Furthermore, HSPA2 and CARM1 were found to form a protein complex and regulated ICM-specific gene expression. Collectively, these results suggest that *Hspa2* expression patterns can bias cell fate in the mouse embryo.

2. Materials and methods

2.1 Antibodies

In this study, we utilized CDX2 (ab76541), NANOG (ab17338) and H3R26me2 (ab127095) antibodies from Abcam; a CARM1 (12495s) antibody from Cell Signaling Technology; SOX2 (11064-1-AP), GAPDH (10494-1-AP), CARM1 (55246-1-AP), HSPA2 (12797-1-AP) and ACTIN (66009-1-Ig) antibodies from Proteintech; and an OCT4 (sc-5279) antibody from Santa Cruz Biotech.

2.2 Mouse embryo collection

Animal experiments were approved by the Ethics Committee of Shandong University. ICR mice (6–8 weeks-of-age) were purchased from Beijing Vital River Laboratory Animal Technology Co., Ltd. and bred under standard SPF conditions with a 12 h dark and 12 h light cycle with food and water provided ad libitum). Superovulation was induced by an intraperitoneal injection of 5 IU of pregnant mare's serum gonadotrophin (PMSG) (110914564, Ningbo Sansheng Biological Technology Co., Ltd) followed 46–48 h later by 5 IU of human chorionic gonadotrophin (HCG) (110911282, Ningbo Sansheng Biological Technology Co., Ltd.). Oocytes were collected 15 h after hCG injection. First, the collected oocytes were fertilized in G-IVF (#10,136, Vitrolife) with epididymal sperm from adult males that had been capacitated in G-IVF for 1 h. All zygotes were cultured in G1 medium (Vitrolife, Sweden) at 37 °C in the presence of 6% CO_2 and 5% O_2. Early 2-, late 2-, 4-, and 8-cell embryos, and morula- and blastocyst stage embryos were collected after 22–26, 34-38, 48–50, 60-65, 70–75, and 96–100 h of culture, respectively. Single blastomeres were isolated from 2- and 4-cell embryos and cultured to the morula stage as described previously[20].

2.3 Cell culture and transfection

Mouse ESCs (mESCs) were cultured on plates coated with 0.1% gelatin (Oricell) under feeder-free conditions with 5% CO2 at 37 °C. For cell transfection, the mESCs were transfected with si-*Hspa2* using Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions. After transfection for 36 hours, cells were collected for further analysis.

2.4 RT-qPCR analysis

For RT-PCR analysis, we used the Single Cell Sequence Specific Amplification Kit (Vazyme, P621-01) and ChamQ Universal SYBR qPCR Master Mix (Vazyme, Q711-03) in accordance with the manufacturer's instructions. Primers for the candidate genes were designed by online Primer 3.0 software based on the sequence data obtained from the NCBI database. Primer sequences are listed in **Table 1**. Gene expression was quantified by the comparative CT method. Values are expressed as the mean ± standard deviation (SD). Expression was normalized to Actin. The relative amount of transcript present in each cDNA sample was calculated using the $2^{-\Delta\Delta CT}$ method. All experiments were repeated at least three times, and statistical analysis was performed on individual experimental sets. All measurements were performed intriplicate for each experiment.

2.5 Automated western immunoblotting

Western immunoblots were performed on a PeggySue (ProteinSimple) system using a Size Separation Master Kit with Split Buffer (12–230 kDa) in accordance with the manufacturer's instructions. Capillary-based immunoassays were performed using the Wes-Simple Western method (Proteintech: HSPA2 (12797-1-AP), 1:15 dilution; GAPDH (10494-1-AP),1:35 dilution; SOX2 (11064-1-AP), 1:20 dilution; Abcam: CDX2 (ab76541), 1:10 dilution; Santa Cruz Biotech: OCT4 (sc-

Gene	Forward primer (5'-3')	Reverse primer (5'-3')
Hspa2	AAGATTTCTTCAACGGCAAGGAG	GGGATCGTGGTGTTTCTCTTGAT
Oct4	GGGTGATGGGTCAGCAG	TCCGCAGAACTCGTATGC
Sox2	AACGCCTTCATGGTATGG	CTCGGTCTCGGACAAAAG
Nanog	CTGCTCCGCTCCATAACT	GGCTTTCCCTAGTGGCTT
Cdx2	GAAGGGGTGGTGGGTTC	AGGTTGGCTCTGGCATTT
Actin	GCTCTTTTCCAGCCTTCCTT	GTACTTGCGCTCAGGAGGAG
Hprt	CGAGATGTGATGAAGGAAATGG	AGCAGGTCAGCAAAGAATTTATAGC

Table 1.

List of mouse primer pairs used for RT-qPCR analysis

5279), 1:15 dilution) and an anti-rabbit or an-mouse detection module (ProteinSimple) [21]. Protein expression was detected by chemiluminescence and quantified by area under the curve (AUC) analysis using the Compass for Simple Western program (ProteinSimple).

2.6 Microinjection and HSPA2 knockdown

Aliquots of 2–5 pl of siRNA at 20 μM concentration were microinjected into the nucleus of zygote or one blastomere of the two-cell embryos using an Eppendorf (Hamburg, Germany) micromanipulator under a Leica inverted microscope, and then cultured in fresh G1 medium.

To knockdown Hspa2, two siRNAs were designed for each gene. The siRNAs of *Hspa2* are GGTGCAACAACTTAGTTTA, GCACTGCAGTGATATTAAA. For microinjection in each KD group,20 μM of each siRNA was used. siRNAs were ordered from RINOBIO.

2.7 Western blot analysis and co-immunoprecipitation (Co-IP)

Protein concentration was determined with a BCA protein assay (Bio-Rad, Hemel Hempstead, UK). Proteins were then separated by 12% SDS-PAGE and transferred to polyvinylidene difluoride membranes (Millipore, Bedford, MA, USA) at 300mA for 90min. First, the membranes were incubated in TBST containing 5% non-fat milk for 2h; then, the membranes were incubated with primary antibodies (Abcam: CDX2 (ab76541), 1:1000 dilution; NANOG (ab17338) 1:800 dilution; Cell Signaling Technology: CARM1 (12495s), 1:1000 dilution; Proteintech: SOX2 (11064-1-AP), 1:800 dilution; GAPDH (10494-1-AP), 1:1500 dilution; HSPA2 (12797-1-AP), 1:1000 dilution; ACTIN (66009-1-Ig), 1:700 dilution; Santa Cruz Biotech: OCT4 (sc-5279), 1:800 dilution) overnight at 4°C. On the second day, the membranes were washed three times with TBST and then incubated with horseradish peroxidase-conjugated secondary antibody (1:2000 dilution; Proteintech) for 1 h.

The Pierce Co-IP kit (Thermo Fisher Scientific, USA) was used for co-IP assays. In brief, mESCs were lysed using IP lysis buffer on ice. Next, the lysate was treated with agarose resin at 4°C for 1 h followed by incubation with antibodies (Proteintech: HSPA2(12797-1-AP), 1:100 dilution; CARM1(55246-1-AP), 1:60 dilution) at 4°C overnight. The captured antigen was then eluted and SDS-PAGE was performed.

2.8 RNA FISH and immunofluorescence

Fluorescence *in situ* hybridization (FISH) was carried out using an RNA-FISH Probe kit (GenePharma, Shanghai), according to the manufacturer's instruction. In brief, embryos were collected and fixed in 4% paraformaldehyde in PBS-DEPC for 15 min at room temperature, permeabilized in 0.1%BufferA (permeabilizing solution) for 15 min at room temperature, and incubated at 37°C in 2× Buffer C for 30 min. The embryos were then dehydrated in 70%, 85%, and 100% EtOH and dried at room temperature for 5 min each time. The embryos were denatured in buffer E (hybridization buffer) containing 10 μM probe at 73°C for 5 min, incubated at 37°C overnight, and washed in 0.1% Buffer F for 5 min to remove non-specific binding. After washing in 2× Buffer C and 1× Buffer C (neutral solution) for 5 min, the nuclei were stained with DAPI for 5 min

To perform immunofluorescence, we removed the zona pellucida with acid Tyrode's solution (Sigma). After that, embryos were fixed in 4% paraformaldehyde in PBS (KGB5001, KeyGEN BioTECH) at room temperature (RT). Embryos were then permeabilized with 1% Triton-100 (T8787, SigmaAldrich) in PBS for 20 min. Embryos were blocked for 30 min at RT on a shaker with a blocking buffer consisting of 1% BSA in PBS containing 0.1% Tween-20 (P9416, Sigma-Aldrich) and 0.01% Triton-100. Next, embryos were incubated with primary antibodies (Abcam: CDX2 (ab76541), 1:150 dilution; H3R26me2 (ab127095), 1:150 dilution; Santa Cruz Biotech: OCT4 (sc-5279), 1:100 dilution) diluted in blocking buffer on a shaker at 4-°C overnight. On the second day, embryos were washed three times with TBST and incubated with secondary antibody (1:400

dilution; Sigma) and DAPI (D3571, 1:400 dilution; Life Technologies) for 30 min. First, embryos were rinsed three times for 10 min in washing buffer on a shaker at RT; then, the embryos were mounted on slides in washing buffer and examined under a laser scanning confocal microscope (Dragonfly, Andor Technology, UK). Fluorescence images were acquired at multiple 1-μ m intervals in the z-axis with the use of a confocal microscope. Multiple images were processed using ImageJ software for intensity measurement and cell counting (*https://imagej.nih.gov/ij/*). The fluorescence intensity was quantified by normalization to DAPI using the built-in ImageJ function. The regions of interest (ROI) were selected using a custom Python script. The intensity was measured on normalized sections using the ImageJ measure function. Data were normalized with respect to background levels. To analyze the ICM/TE cell numbers, OCT4 and CDX2 signal at inner layers and outer layers were used to distinguish the ICM and TE cells.

2.9 Molecular docking analysis

For molecular docking analysis, we first generated a PPI network and identified core targets. The proposed protein model was based on the crystal structure downloaded from the AlphaFold Protein Structure Database (*https://alphafold.ebi.ac.uk/*) and the two- and three-dimensional (2D/3D) structures of potential active compounds were acquired from PubChem (*https://pubchem.ncbi.nlm.nih.gov/*). Then, molecular docking analysis was performed and visualized in Maestro11.9; the resultant docking score was saved and then compared.

2.10 Proteomics and bioinformatic analysis

For proteomics and bioinformatic analysis, 4 embryos per sample were collected. Four replicates were assessed for NC group and two replicates were assessed for KD group. Protein samples were sonicated three times on ice using a high intensity ultrasonic processor (Scientz) in lysis buffer (8 M urea, 1% protease inhibitor cocktail). Inhibitors were also added to lysis buffer for PTM experiments. The remaining debris was removed by centrifugation at 12,000 g at 4°C for 10 min. The protein samples were then diluted by adding 200 mM of TEAB to urea concentrations < 2 M. Finally, trypsin was added at a 1:50 trypsin-to-protein mass ratio for the first digestion overnight and a 1:100 trypsin-to-protein mass ratio for a second 4 h-digestion. Finally, the peptides were desalted with a Strata X SPE column. The eluted peptides were then cleaned with C18 ZipTips (Millipore) in accordance with the manufacturer's instructions, followed by analysis with LC–MS/MS. The resulting MS/MS data were processed by MaxQuant with an integrated Andromeda search engine (version 1.4.1.2). False discovery rate thresholds for proteins, peptides and modification sites were specified at 1%.

COG, full name is Clusters of Orthologous Groups of proteins. The proteins that make up each COG are assumed to be derived from an ancestral protein, Orthologs are proteins that are derived from different species, evolved from vertical lineages (species formation), and typically retain the same function as the original protein. COG means "cluster of homologous proteins" in Chinese. COGs are divided into two categories, one for prokaryotes and the other for eukaryotes. Those for prokaryotes are generally referred to as COG databases; those for eukaryotes are generally referred to as KOG databases. Compared to other databases, such as NCBI's COG database, EggNOG provides a more comprehensive classification of species and more homologous protein sequences, with phylogenetic tree construction and functional annotation for each homologous gene cluster.

2.11 RNA-seq analysis

For RNA sequencing, 3 embryos per sample were collected. Three replicates were assessed for NC group and three replicates were assessed for KD group. After being denuded of the zona pellucida using EmbryoMax Acidic Tyrode's Solution (MR-004-D, Sigma-Aldrich), RNA-seq libraries were generated following the Smart-seq2 protocol as described previously[22]. Sequencing libraries were generated using TruePep™DNA Library Prep Kit V2 for Illumina® (TD502, Vazyme) according to manufacturer's recommendations. The libraries were performed on an Illumina Hiseq X-ten platform with 150 bp paired-end.

Filtering out low quality and joint sequences, the clean reads of each sample obtained were mapped to the reference genome downloaded from the genome website (*https://ftp.ensembl.org/pub/release-105/fasta/mus_musculus/dna/Mus_musculus.GRCm39.dna.primary_assembly.fa.gz*) using Hisat2. The expression values for each gene were calculated as Fragments Per Kilobase of exon model per Million mapped fragments (TPM+1). DEG analysis was conducted with edgeR package. In this project, the gene were identified as differentially expressed genes (DEGs) if |log2FC| >1 and P<0.05 (FC: fold change). Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were s carried out for all the DEGs, utilizing the "cluster Profiler" package.

2.12 Statistical analysis

Statistical analyzes were performed by GraphPad Prism software (version 9.3, GraphPad Prism Software Inc., San Diego, CA) and the Statistics Package for Social Science (SPSS 11.5; SPSS Inc., Chicago, IL, USA). Statistical significance (**Figures 1E-J**, **2G,I-M**, **3B**,**3E**,**4B-D**,**4F**,**5B-E**,**5H**, **S1C-J**, **S2G**) were calculated with two tailed Student's t test. Levels of significance (**Figures 1C**) were calculated with t test. At least three biological replicates were performed for each experiment and results are represented as mean ± standard error of mean (SEM). p< 0.05 was considered statistically significant.

2.13 Data access

The RNA-sequencing data used in this study were obtained from the NCBI Gene Expression Omnibus (GEO; *http://www.ncbi.nlm.nih.gov/geo/*) under accession number (GSE266025). Proteomic data were available via ProteomeX change with identifier PXD052193.

3. Results

3.1 *Hspa*2 is asymmetrically expressed between late two- and four-cell blastomeres

In our previous study, we established a culture system that allows blastomeres without their zona pellucida to develop in vitro until the blastocyst stage [20]. Previous studies have mainly focused on gene heterogeneity in 4-cell stage mouse embryos[5, 6]. Here, in order to expand the heterogeneity between blastomeres and identify factors that may involve in cell lineage differentiation, single blastomeres from the mouse 4-cell embryo were separated and cultured to the morula stage. By using Smart-seq2 technology, we analyzed the gene expression pattern of the morulae, which developed from the 4-cell blastomeres (**Figure 1A**). *Hspa2* was identified by differential gene expression analysis and showed distinct mRNA levels among the four morulae (**Figure 1B**). In addition, the level of *Nanog* and *Hspa2* mRNA showed stronger positive correlation (R=0.624) in gene expression than random gene pairs (R=0.029), indicating that *Nanog* expression levels increased with the increase of *Hspa2* (**Figure 1C**). When the late 2-cell embryo was divided into single blastomeres and cultured to the morula stage, we obtained similar results that *Hspa2* mRNA distinctly distributed between the two morulae (**Figure S1A, B**). Next, we investigated the expression of *Hspa2* during preimplantation embryo development from meiosis II to the blastocyst stage. qRT-PCR analysis showed that *Hspa2* mRNA was stored maternally at the MII oocyte and decreased at the two-cell stage but remarkably increased from the 2-cell to the 4-

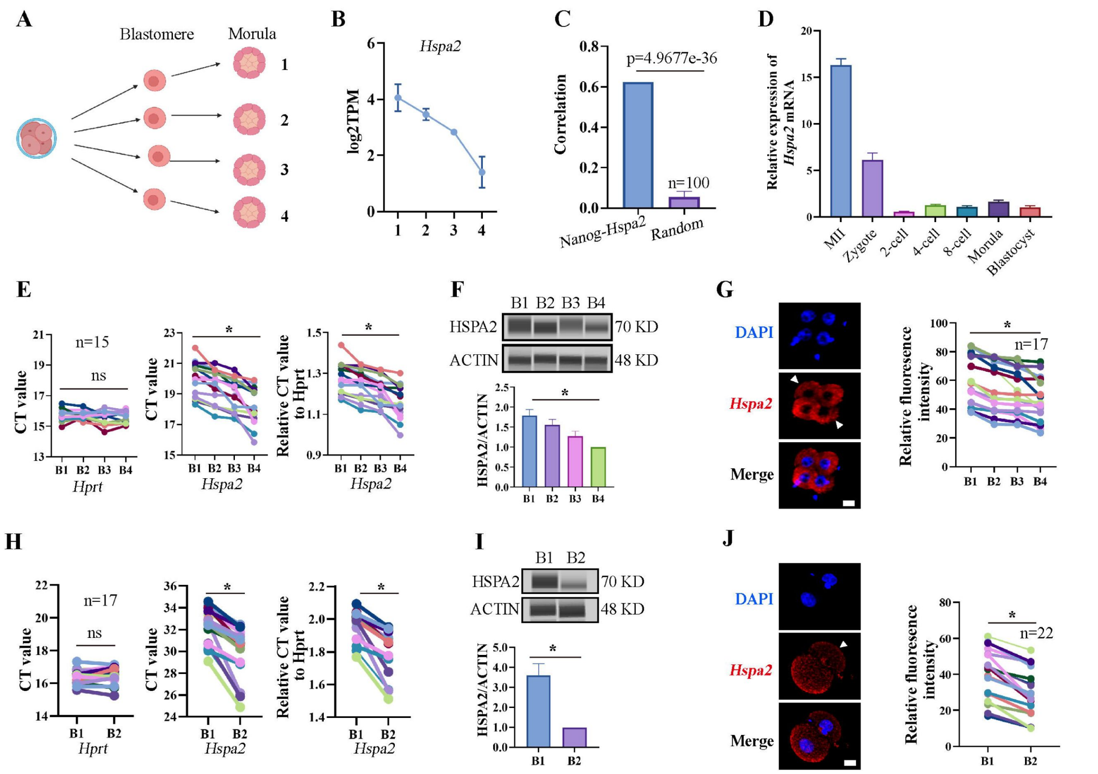


Figure 1.

Hspa2 is asymmetrically expressed between mouse 2- and 4-cell blastomeres.

(A) Schematic overview. The 4-cell embryo divided into single blastomeres and cultured to the morula stage. Created with biorender.com. **(B)** *Hspa2* expression is heterogeneous in morula developed from 4-cell blastomeres. Three experimental replicates were performed. **(C)** *Nanog* and *Hspa2* show stronger positive correlation (R=0.624) in gene expression than random gene pairs (n=100, n means the number of random gene pairs). (D) qPCR analysis of the relative expression of Hspa2 mRNA form the meiosis II to blastocyst stages. The error bars represent SEM. About 20 embryos of each stage were used and three experimental replicates were performed. **(E and H)** Single-cell qRT-PCR results showing *Hspa2* distribution between 4- (E) (n=15), late 2- (H) (n=17) cell blastomere. *Hprt* as the housekeeping gene. CT values are used for level analysis. Two-tailed Student's t tests were used for the statistical analysis. Blastomeres are named B1 to B4 (E) or B1 and B2 (H) according to increasing *Hspa2* concentration. **p* <0.05. **(F and I)** Automated western immunoblotting results showing HSPA2 distribution between 4- (F) and late 2- (I) cell blastomeres. Blastomeres are named B1 to B4 (E) or B1 and B2 (H) according to decreasing *Hspa2* concentration. The band intensity was assessed. There experimental replicates were performed. **p* <0.05. **(G and J)** FISH of *Hspa2* mRNA in early 4- (G) (n=17), late 2- (J) (n=22) cell embryos. Blastomeres with low expression as white triangles. RNA-FISH fluorescence quantified and normalized to the nucleus with the strongest staining per individual two- or four-cell embryo. Blastomeres are named B1 to B4 (G) or B1 and B2 (J) according to increasing *Hspa2* concentration. Three experimental replicates were performed. Bar, 25 µm. **p* <0.05.

cell stage, coinciding with zygotic genome activation (ZGA). There was a slight change in expression as embryos developed to the 8-cell and morula stages, and developed into blastocysts that had higher expression levels than before the 2-cell stage (**Figure 1D**).

To directly examine the heterogeneity among individual blastomeres, we separated 4-cell embryos into individual four blastomeres which were then analyzed by single-cell qRT-PCR and Jess TM Simple Western System based automated western immunoblotting with high sensitivity. The results showed that the mRNA and protein levels of HSPA2 differed among the four blastomeres (**Figure 1E, F**), and confirmed by fluorescence *in situ* hybridization (FISH) analysis (**Figure 1G**). We also identified heterogeneous HSPA2 expression during the late 2-cell stage (**Figure 1H-J**). For the early 2-cell stage, however, HSPA2 was symmetrically distributed between the two blastomeres (**Figure S1C-E**). These results indicating that the heterogeneity occurred as early as the late 2-cell stage. Thus, we hypothesized that the heterogeneous distribution of HSPA2 in late 2-cell plays a direct or indirect role in the lineage commitment of early embryonic stages.

3.2 The reduced expression of *Hspa2* leads to the reduced expression of ICM-marker

To test our hypothesis, we knocked down the *Hspa2* mRNA level by injecting *Hspa2*-specific small interfering RNA (siRNA) into the cytoplasm of zygote, and NC-FAM (negative control with FAM label) siRNA-injected embryos served as the control (**Figure 2A**). The differentially expressed genes were detected by RNA-seq for *Hspa2* siRNA (Knockdown; KD) and control (NC) groups at the 8-cell stage (**Figure 2B**). In total, 96 up-regulated genes and 410 down-regulated genes were identified in the *Hspa2*-KD group when compared to the NC group (**Figure 2C, D**). To investigate the potential biological functions of the DEGs, we conducted GO and KEGG enrichment analysis. The most significant GO enriched terms were closely related to cell fate commitment, stem cell population maintenance, stem cell differentiation and blastocyst development (**Figure 2E**). According to KEGG analysis, the down-regulated genes were primarily associated with Hippo and Wnt signaling pathways (**Figure 2F**), which are known to play an important role in cell proliferation, differentiation, and fate decisions in embryonic development[23–26]. Furthermore, the blastocyst formation rate in the *Hspa2-KD* group (52.3±2.5%) was significantly lower than that in the NC group (81.6±2.0%) (**Figure 2G, H**), thus suggesting that HSPA2 plays an essential role in preimplantation embryonic development. In addition, we found that the down-regulation of *Hspa2* resulted in a remarkable down-regulation of specific ICM-marker genes (*Oct4*, *Sox2* and *Nanog*) at the 4-cell and blastocyst stages. Meanwhile, the mRNA levels of a TE-marker gene, Cdx2, did not significantly change between the groups (**Figure 2I, J**). We then used automated Western immunoblotting to detect the protein levels of ICM and TE markers in Hspa2-KD embryos, and found that the expression levels of ICM-marker proteins (OCT4 and SOX2) were inhibited when HSPA2 was down-regulated at the 4-cell and blastocyst stages, while the TE-marker protein CDX2 was not significantly different between the groups (**Figure 2K, L**). Similarly, for mouse embryonic stem cells (mESCs), the knockdown of HSPA2 resulted in a significant reduction for the ICM-marker proteins, and no change for CDX2 (**Figure 2M**). These results suggest that HSPA2 plays a role in the regulation of ICM-specific gene expression.

3.3 The knockdown of *Hspa2* expression prevents blastomeres from an ICM fate

To investigate the functional role of HSPA2 in the first lineage segregation during early embryonic development, we examined the proportion of ICM and TE after Hspa2-KD. It was showed that the number of ICM and total cells in the blastocyst were significantly decreased, but the number of TE cells did not change significantly (**Figure 3A, B**). We then randomly co-injected green fluorescent protein (*Gfp*) mRNA with either *Hspa2*-siRNA or NC-FAM into one blastomere of the 2-cell embryos to trace the fate of its descendent cells into the blastocyst (**Figure 3C**). Knock-down

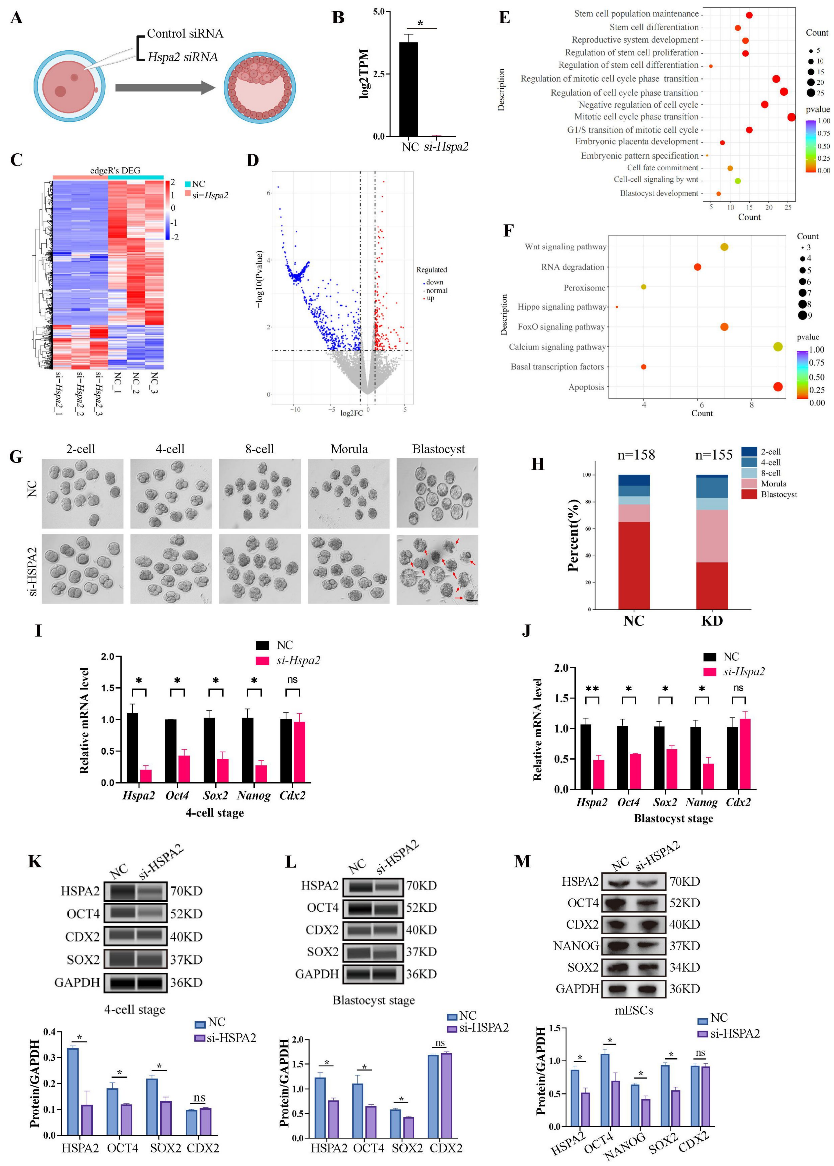


Figure 2.

Knockdown of *Hspa2* lead to decreased expression of ICM-mark at both the mRNA and protein level.

(A) Schematic diagram of microinjection of zygotic embryos. Created with biorender.com. **(B)** Bar charts showing the expression of *Hspa2* in the negative control (NC) group (injected with negative control siRNA) and siRNA targeting group based on RNA-seq. The error bars denote the standard deviations of three experimental replicates of RNA-seq. *p <0.05. **(C)** Heat map of the hierarchical clustering analysis of the DEGs between NC and KD groups. **(D)** Volcano plots of significantly differentially expressed genes (DEGs). Red, upregulated; Blue, down-regulated. **(E)** GO enrichment analysis from down-regulated DEGs between NC and KD groups. **(F)** Significantly down-regulated DEGs were analyzed by KEGG enrichment. **(G)** Representative images of preimplantation embryos in the NC and KD groups from 2-cell to blastocyst. Red arrowhead denotes degenerating blastocysts. Bar, 100 μm. **(H)** Bar plots showing the developmental rates of NC group and KD group at the blastocyst stage.. Three experimental replicates were performed. **(I and J)** qRT-PCR of 4-cell embryo and blastocyst showed that ICM-marker gene (*Oct4*, *Sox2* and *Nanog*) mRNA was significantly lower in the interference compared with the NC group. There were no significant differences in Cdx2 between the two groups. *p < 0.05, **p < 0.01. **(K and L)** Automated western blotting analysis of ICM-marker protein (OCT4, SOX2 and NANOG) expression in the interference and NC groups. Three experimental replicates were performed. *p <0.05. **(M)** Western blot analysis of ICM-marker protein (OCT4, SOX2 and NANOG) and TE-marker protein (CDX2) expression in mESCs. The band intensity was assessed with ImageJ. Three experimental replicates were performed. *p <0.05.

Hspa2 in one blastomere at the 2-cell stage also led to developmental defects (**Figure S2A, B**). Compared with the GFP+NC group (48.2%±7.4%), the GFP+*Hspa2*-KD group (33.8 %±4.3%) exhibited a significantly reduced ratio of progeny cells with an ICM fate. Moreover, the number of ICM-marker OCT4 positive cells was found to be significantly reduced in the GFP+*Hspa2*-KD group (27.1±3.8%) than in the GFP+NC group (37.0%±2.5%) (**Figure 3D, E**). This is consistent with our GO enriched terms of RNA-Seq in that the negative regulation of cell cycle, G1/S transition of mitotic cell cycle, mitotic cell cycle phase transition and regulation of mitotic cell cycle phase transition was found in *Hspa2*-KD group (**Figure 2E**). Collectively, these results suggested that HSPA2 has a role in ICM lineage establishment and its deficiency leads to partial defects in the ICM cells.

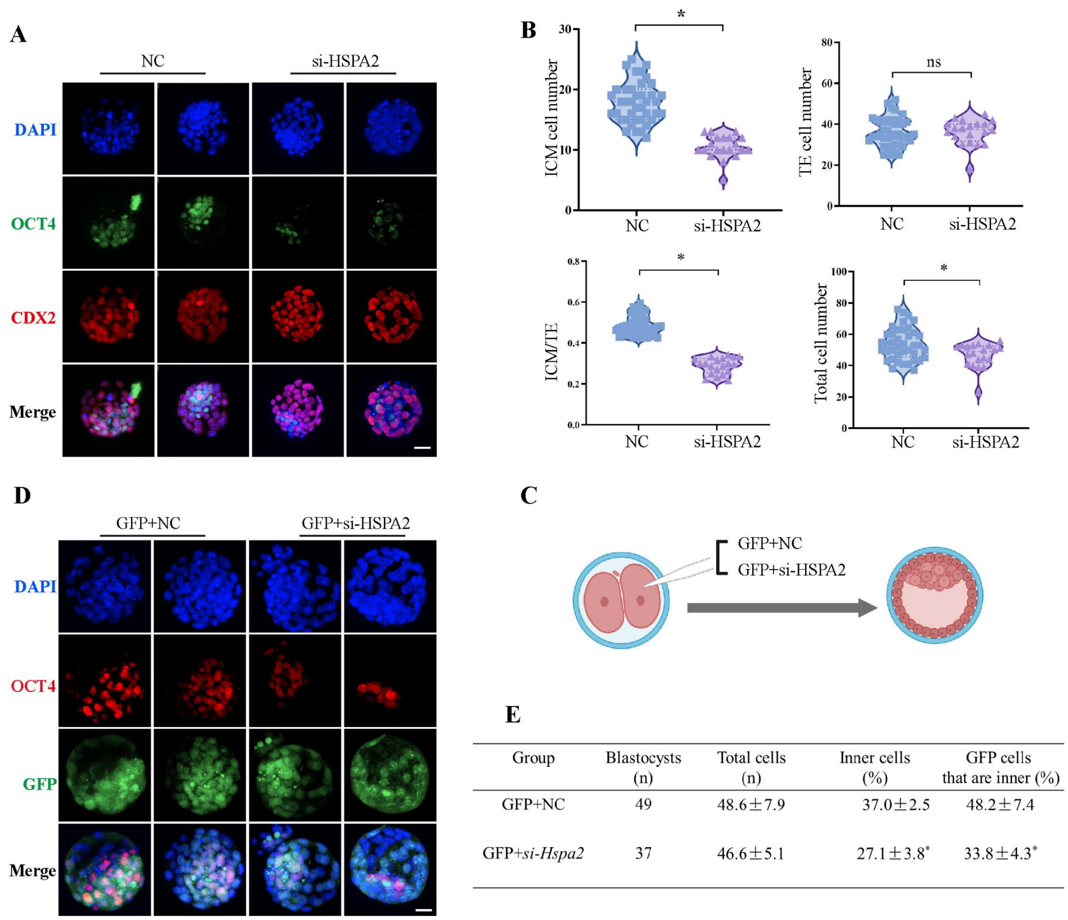
3.4 The overexpression of *Hspa2* does not induce blastomere cells to bias an ICM fate

To determine whether the redundant level of HSPA2 protein also affected first lineage segregation, we overexpressed *Hspa2* by microinjecting *Hspa2* mRNA (approximately 400ng/μL) into the zygote. Western blotting assays showed that the HSPA2 was overexpressed in the blastocyst stage (**Figure S3**). We observed that the blastocyst formation rate was not affected significantly (**Figure 4A, B**), thus showing that the overexpression of *Hspa2* did not exert adverse effects on embryonic development. In addition, we verified that the mRNA levels of the ICM-marker genes (*Oct4, Sox2* and *Nanog*) and the TE-marker gene *Cdx2* at the 4-cell and blastocyst stage did not differ significantly (**Figure 4C, D**). We also examined the number of ICM cells and TE cells by immunostaining OCT4 and CDX2 in blastocysts from the overexpression and control groups and found they were not altered significantly (**Figure 4E, F**). To trace the fate of the descendent, we co-injected GFP mRNA along with either *Hspa2* or NC mRNA randomly into one blastomere of 2-cell embryos and counted the proportion of ICM and TE cells of blastocysts. We found that the percentage of GFP cells that were present in the ICM at the blastocyst stage was similar between the two groups (48.2±7.4 vs. 47.3±4.4), and the percentage of ICM cells also did not differ significantly (37.0±2.5 vs. 36.2±1.9) (**Figure 4G, H**). These observations indicated that the redundant level of HSPA2 were not sufficient to induce blastomere cells to bias an ICM fate.

3.5 HSPA2 interacts with CARM1 and alters H3R26me2 levels

To further investigate the mechanism underlying the function of HSPA2 in cell fate decision during early embryonic development, we next analyzed differentially expressed proteins (DEPs) at the blastocyst stage in the *Hspa2*-KD and NC groups by applying untargeted proteomics. Volcano map screening identified a total of 43 annotated DEPs, of which 15 proteins were significantly up-regulated and 28 proteins were downregulated (**Figure S4A**). These DEPs were mainly involved in post-translational modification by using Clusters of Orthologous Groups (COG)/Karyotic Orthologous Groups (KOG) functional annotation statistics (**Figure S4B**). In order to compare the functional similarities and differences between proteins with different differential multiples, we divided the DEPs into four classifications: Q1 (<0.5), Q2 (0.5-0.667), Q3 (1.5-2.0), and Q4 (>2.0) according to differential expression multiples (**Figure S4C**). Biological processes analysis showed that DEPs were associated with positive regulation of the Wnt signaling pathway and cellular development (**Figure S4D**). In terms of the cellular component, the DEPs were related to the methylome and methyltransferase complex (**Figure S4E**). These results indicated that the DEPs were mainly associated with the post-translational modification of proteins and cell fate decision.

Coactivator-associated arginine methyltransferase 1 (CARM1), also known as PRMT4, was the first protein arginine methyltransferase which was proved to be associated with transcriptional activation by methylating and/or regulating histone H3 and non-histone proteins, such as coactivators and transcription factors [27, 28]. Recent studies have demonstrated that CARM1



Group	Blastocysts (n)	Total cells (n)	Inner cells (%)	GFP cells that are inner (%)
GFP+NC	49	48.6±7.9	37.0±2.5	48.2±7.4
GFP+*si-Hspa2*	37	46.6±5.1	27.1±3.8*	33.8±4.3*

Figure 3.

Knockdown of *Hspa2* prevents blastomeres from undergoing an ICM fate.

(A) Immunofluorescence of OCT4 (green) and CDX2 (red) in NC (n=40) and si-*Hspa2* (n=36) groups. OCT4 was used as an ICM marker. Nuclei were visualized with DAPI staining (blue). Bar, 50 μm. **(B)** The number of ICM and TE. OCT4 was used as an ICM marker and CDX2 was used as a TE marker. *p <0.05. **(C)** Schematic diagram of microinjection of early 2-cell embryo single blastomeres. Created with biorender.com. **(D)**OCT4 and GFP fluorescent staining of blastocysts. OCT4 was used as an ICM marker. The results show that less OCT4-positive cells are GFP-positive in embryos injected with GFP and *Hspa2 siRNA* (GFP+*si-Hspa2*). Three experimental replicates were performed. Bar, 50 μm. **(E)** Analysis of the distribution of progeny of injected blastomere at the blastocyst stage based on fluorescence staining. Total cells (n) are the total number of cells in the blastocyst; inner cells (%) is the percentage of inner cells relative to the total number of cells in the blastocyst; GFP cells that are inner (%) is the percentage of GFP-positive inner cells relative to the total number of GFP-positive cells in the blastocyst. Data derived from three independent experiments and presented as mean ± standard deviation. Three experimental replicates were performed. *p <0.05.

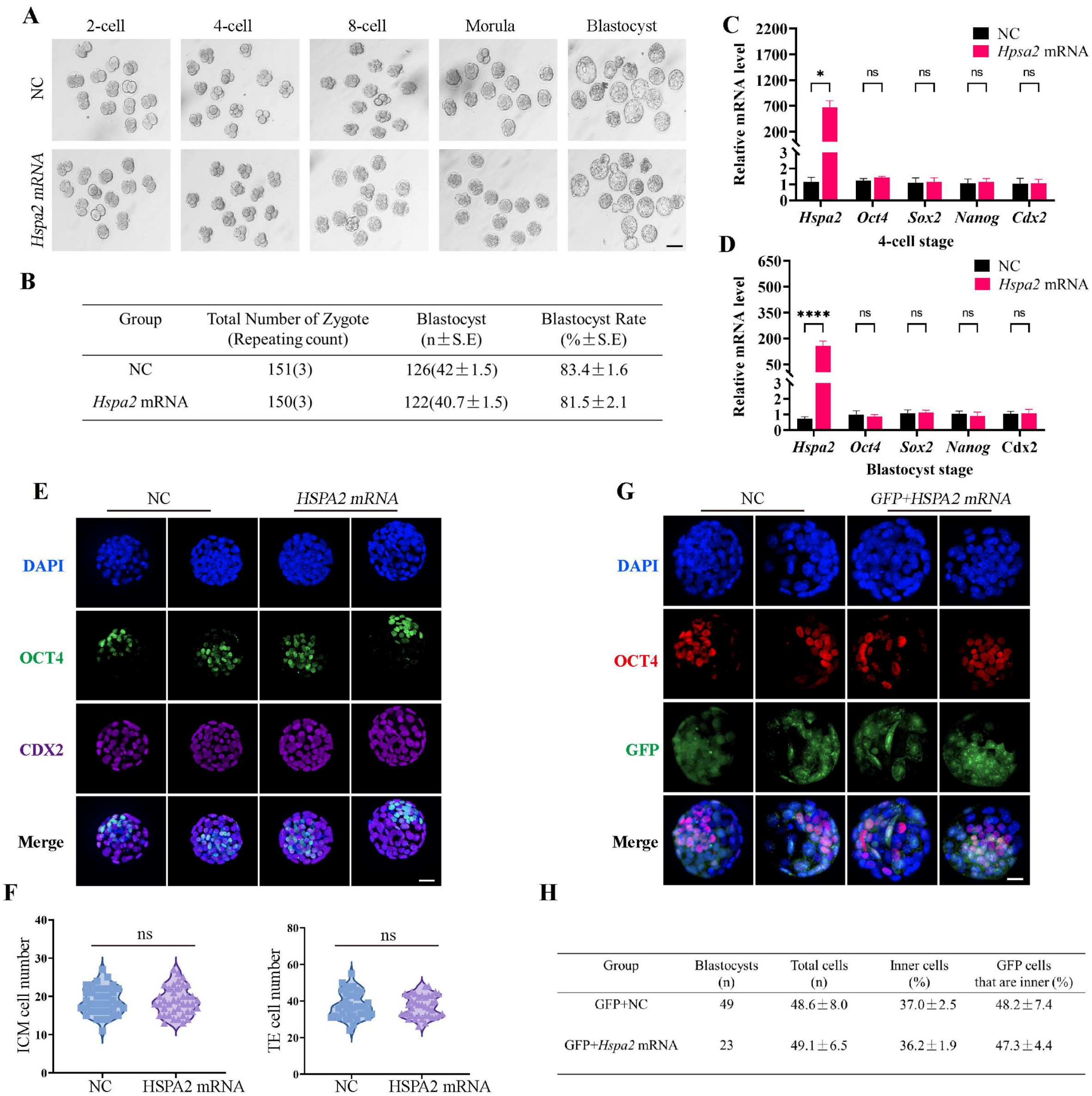


Figure 4.

Overexpression of *Hspa2* not induces blastocyst bias towards ICM fate.

(A) Representative images of preimplantation embryos in the NC and *Hspa2* mRNA groups from 2-cell to blastocyst. Bar, 100 µm. **(B)** The blastocyst rate between NC (n=151) and overexpression (n=150) groups. Three experimental replicates were performed. **(C and D)** qRT-PCR of 4-cell embryo and blastocyst showed that ICM-marker gene (*Oct4, Sox2* and *Nanog*) and TE-marker gene (*Cdx2*) were no significant differences in the overexpression compared with the control group. Three experimental replicates were performed. *p <0.05, ****p <0.001. **(E)** Immunofluorescence of OCT4 (green) and CDX2 (purple) in NC (n=38) and overexpression (n=40) groups. OCT4 was used as an ICM marker. Nuclei were visualized with DAPI staining (blue). Bar, 50 µm. **(F)** The number of ICM and TE. OCT4 was used as an ICM marker and CDX2 was used as a TE marker. **(G)** OCT4 and GFP fluorescent staining of blastocysts. OCT4 was used as an ICM marker. The results show that the percentage of GFP cells that were present in the ICM was similar between the two groups. Three experimental replicates were performed. Three experimental replicates were performed. Bar, 50 µm. **(H)** Analysis of the distribution of progeny of injected blastomere at the blastocyst stage based on fluorescence staining. Total cells (n) are the total number of cells in the blastocyst; inner cells (%) is the percentage of inner cells relative to the total number of cells in the blastocyst; GFP cells that are inner (%) is the percentage of GFP-positive inner cells relative to the total number of GFP-positive cells in the blastocyst. Data derived from three independent experiments and presented as mean ± standard deviation. Three experimental replicates were performed.

regulates early mouse development and plays a role in the determination of cell fate [29, 30]. CARM1 levels, along with its specific histone marks (H3R17me2a and H3R26me2a), were found to vary among the four-cell blastomeres and that high levels of CARM1 led to increased levels of H3R26me. This varied distribution led to biased subsequent fate of the blastomeres towards the ICM [6, 31]. Next, we considered whether HSPA2 regulates the decision of the first cell-fate through CARM1. Molecular docking revealed that HSPA2 docked efficiently with CARM1 via several H-bonds and pi-interactions, with strong binding affinity (**Figure 5A**). The Search Tool for the Retrieval of Interacting Genes (STRING) database was used to predict interactions between HSPA2 and CARM1. The results showed that HSPA2, CARM1 and ICM-specific proteins interacted with each other (**Figure S4F**). In addition, we found that the *Hspa2*-KD led to reduced CARM1 mRNA as well as protein levels at the 4-cell and blastocyst stages (**Figure 5B-E**). The knockdown of HSPA2 in mESCs also caused a significant reduction in the levels of CARM1 (**Figure S4G**).

To further investigate whether HSPA2 could interact with CARM1, we performed co-immunoprecipitation (co-IP) assays in mESCs using HSPA2 and CARM1 as bait. Co-IP analysis showed that HSPA2 and CARM1 proteins interacted effectively (**Figure 5F, G**). Moreover, we investigated whether HSPA2 could regulate H3R26me2 modification, a known event that occurs downstream of CARM1 [6]. Immunofluorescence staining confirmed that CARM1-mediated H3R26me2 levels were significantly reduced in the interference group at the 4-cell stage (**Figure 5H**). Collectively, our data demonstrated that HSPA2 involved in the segregation of embryonic cell lineage by forming a complex with CARM1 and through H3R26me2 modification.

Discussion

During the development of the fertilized eggs to blastocyst stage, the fundamental questions are to explain how the totipotent fertilized egg develops into different cell types and when cellular heterogeneity occurs. Previous studies demonstrated that molecular polarity and differential gene regulation by signaling pathways drive the first cell-fate decision in mammals [26, 32–34]. The tracing of embryonic lineages from the 2-cell stage onwards by live embryo labeling demonstrated that the cellular heterogeneity is initiated early in the embryo, and thus was proposed as cellular heterogeneity hypothesis [35]. H3R26me and its methyltransferase CARM1exhibit asymmetric expression patterns in the 4-cell stage of mouse embryos and plays an important role in regulating the first cell-fate decision. [36, 37]. PRDM14 is also heterogeneously expressed in 4-cell stage mouse embryos and interacts with CARM1 to drive progenies towards pluripotent ICM fate by increasing the expression levels of H3R26me in 4-cell embryos [38]. Despite the progress in our understanding of the mechanisms governing early cell fate decisions, many important questions still remain unanswered. We do not yet fully comprehend when the initial heterogeneities among cells arise and how are they translated into divergent patterns of gene expression [39, 40].

Although two-cell blastomeres are generally considered to be totipotent, the previous reports showing that when two-cell blastomeres are separated, in the majority of cases, only one of the two-cell blastomeres develops into a mouse [41]. In this study, we found that the expression levels of HSPA2 were symmetrically distributed between the two blastomeres at the early 2-cell stage, but showed heterogeneous distribution at the late 2-cell embryos. Interestingly, Hspa2 mRNA also heterogeneous distribution at the late 2-cell and 4-cell embryos after microinjection of siHspa2 in zygotes (**Figure S1 F-G**). This indicates that the heterogeneity of HSPA2 at the late 2-cell embryo may be initiated concomitant with the zygotic genome activation (ZGA) [7]. Alternatively, the asymmetry may be triggered by unequally distributed unknown factors that already existed in the early 2-cell embryos or the heterogeneous spatial distribution of maternal genes in the zygote.

The reduced expression of HSPA2 led to the reduced expression levels of ICM-marker genes at both the mRNA and protein levels. Interfering with HSPA2 expression in one blastomere of the 2-cell embryos reduced the contribution of that blastomere to ICM differentiation. ICM cells were

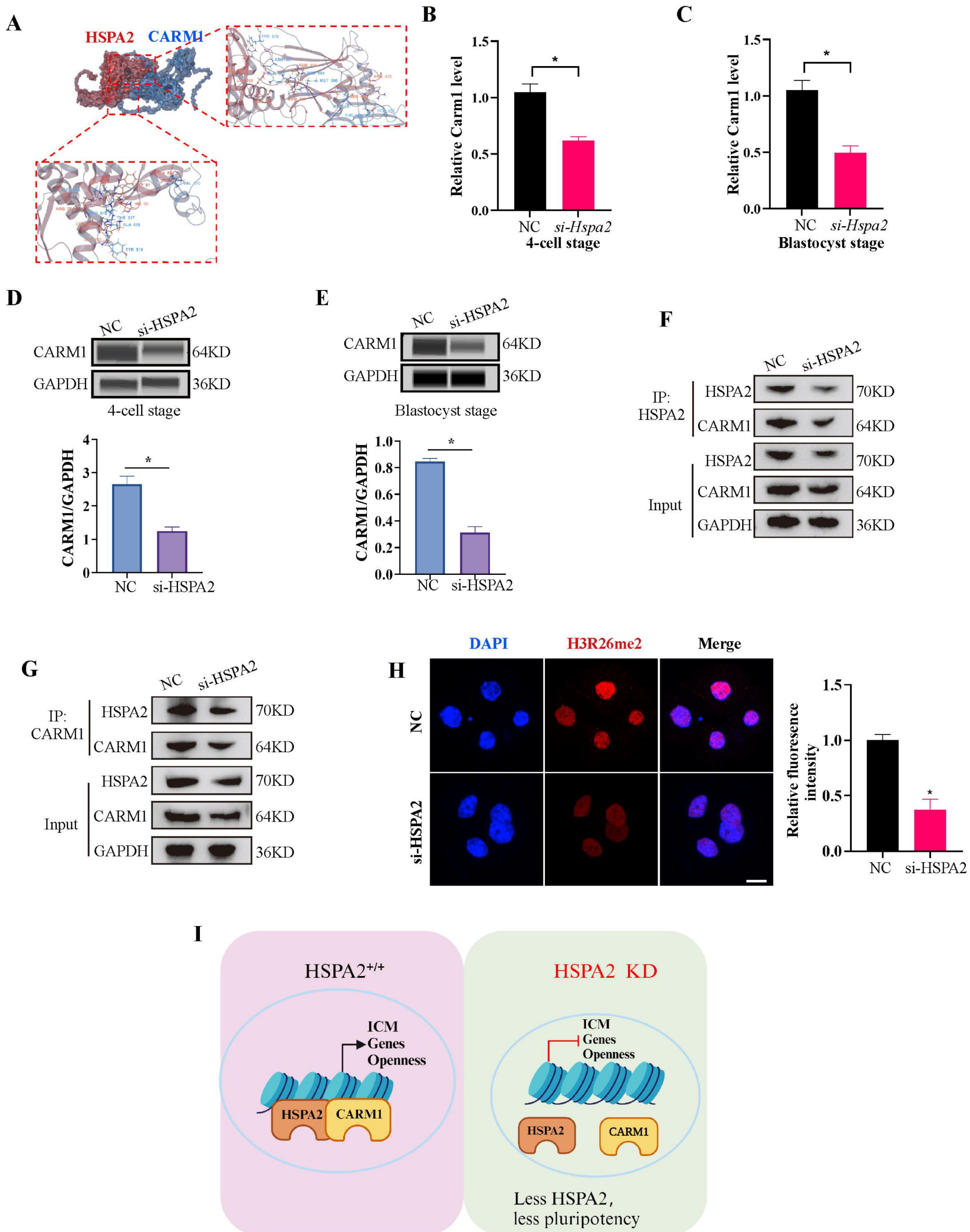


Figure 5.

HSPA2 and CARM1 form a complex.

(A) Molecular docking analysis for HSPA2 binding to CARM1 proteins. **(B and C)** qRT-PCR of 4-cell embryo and blastocyst showed that *Carm1* was significantly lower in the interference compared with the control group. Three experimental replicates were performed. *$p < 0.05$. **(D and E)** Automated western blotting analysis of CARM1 expression in the interference and control groups. The band intensity was assessed. Three experimental replicates were performed. *$p < 0.05$ **(F and G)** CO-IP analysis of HSPA2 and CARM1 in mESCs. **(H)** H3R26me2 staining of 4-cell embryo shows that HSPA2 knockdown led to a dramatic decrease in H3R26me2 modification. Intensity relative to DAPI signal was used. Three experimental replicates were performed. Bar, 50 µm.*$p < 0.05$. (I) The model of HSPA2 physically binds to CARM1 to activate ICM-specific gene expression.

still formed due to the incomplete knockdown achieved or the possibility that redundant pathways exist. Notably, the number of ICM and total cells in the blastocyst was significantly reduced when compared to the control group (**Figure 3B, E**). We observed that the GO enriched terms were also closely related to negative regulation of cell cycle, G1/S transition of mitotic cell cycle, mitotic cell cycle phase transition and regulation of mitotic cell cycle phase transition (**Figure 2E**). HSPA2 knockdown via siRNA reduced cell proliferation and led to G1/S phase cell cycle arrest [42]. Thus, we guessed that knockdown of HSPA2 results in lower cell number in ICM by interfering with the cell cycle. Additionally, the lower cell number in ICM may also associated with failed cytokinesis and the generation of binucleated or polyploid cells. Interestingly, half of the ICM cells still form with HSPA2 deficiency (**Figure 3B**), possibly due to the incomplete knockdown achieved or the possibility that redundant pathways exist.

However, the overexpression of HSPA2 did neither adversely affect embryo development and nor induce blastomere bias towards an ICM fate. It may be because that once the intracellular concentration of *Hspa2* reaches a threshold level, it was not able to control the target gene expression. Furthermore, we also found that HSPA2 interacted with CARM1, and reduced HSPA2 level, led to the reduced expression of CARM1, and thus reduced the levels of H3R26me2 modification at the 4-cell stage. Collectively, these results suggest that HSPA2 physically binds to CARM1 to activate ICM-specific gene expression through H3R26me2 modification (**Figure 5I**).

In fact, our RNA-seq results revealed that the differentially expressed genes (DEGs) at blastocyst stage with HSPA2 knocking down were also enriched in Wnt signaling pathway and Hippo signaling pathway(**Figure 2E**), which are also reported to be involved in cell proliferation, cell polarity, and cell fate determination during embryonic development. [26] [43]. This indicates that several alternative mechanisms are compatible with the asymmetric hypothesis, including polarized cell division and differential gene regulation by signaling.

In summary, our findings provide an exciting step forward for the first time that the role of HSPA2 in governing the first cell-fate decision in the mouse embryo lineage allocation. It may be beneficial for achieving a better understanding of embryogenesis patterns and cell differentiation.

Supplemental Figure caption

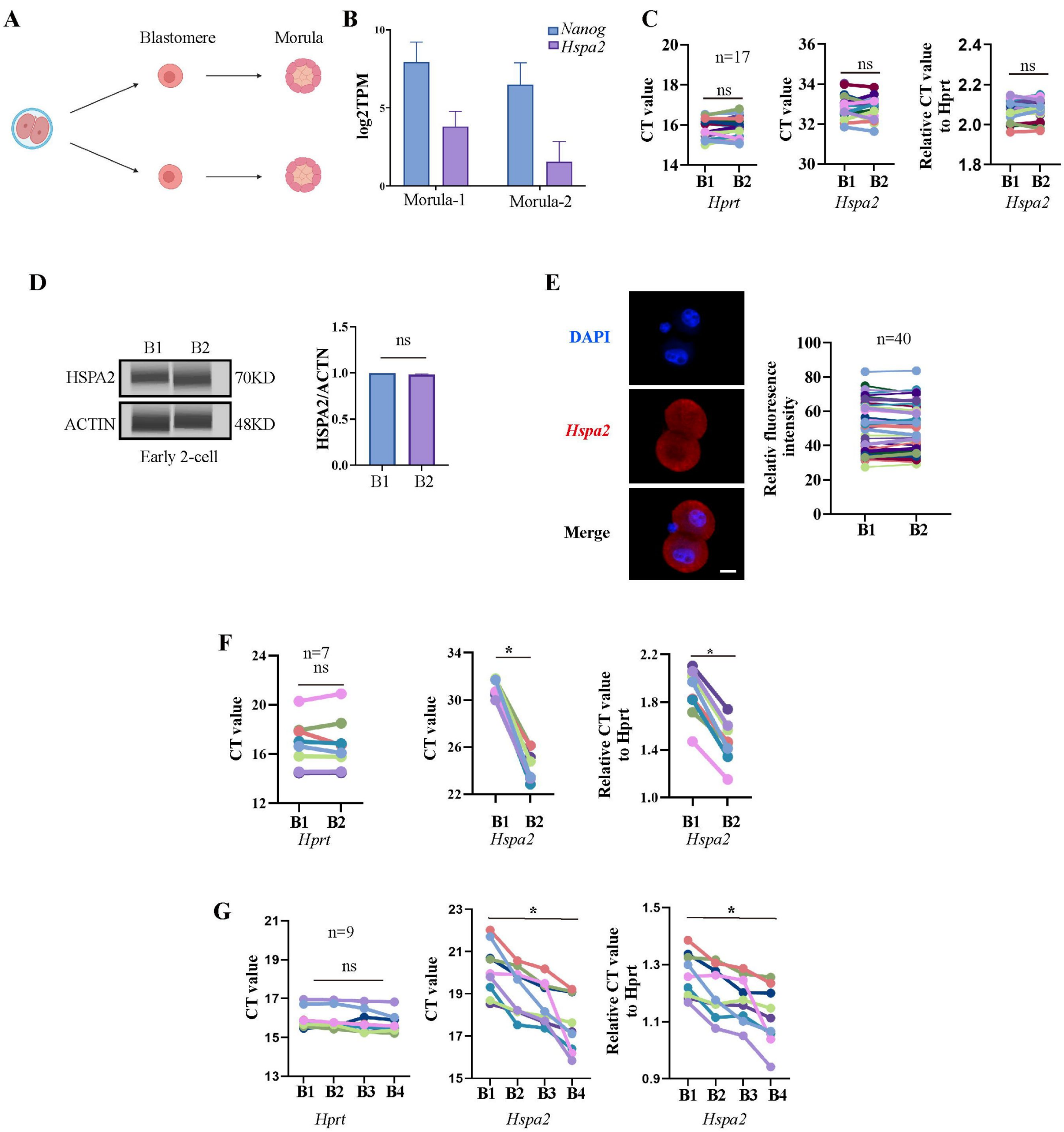


Figure S1.

Hspa2 is asymmetrically expressed between mouse 2- and 4-cell blastomeres, Related to Figure 1.

(A) Schematic overview. Late 2-cell divided into single blastomeres and cultured to the morula stage. Created with biorender.com. **(B)** *Hspa2* expression is heterogeneous and the level of *Nanog* is positively correlated with the expression of *Hspa2* in morula developed from late 2-cell blastomeres. Three experimental replicates were performed. **(C)** Single-cell qRT-PCR results showing *Hspa2* distribution between early 2-cell (n=17) blastomere. *Hprt* as the housekeeping gene. CT values are used for level analysis. Two-tailed Student's t tests were used for the statistical analysis. Three experimental replicates were performed. **(D)** Automated western immunoblotting results showing HSPA2 distribution between early 2-cell blastomeres. Blastomeres are abbreviated named B. The protein level is normalized B1. Three experimental replicates were performed. **(E)** FISH of *Hspa2* mRNA in early 2-cell embryos (n=40). RNA-FISH fluorescence quantified and normalized to the nucleus with the strongest staining per individual early 2-cell embryo. Three experimental replicates were performed. Bar, 25 μm. **(F and G)** Single-cell qRT-PCR results showing *Hspa2* level in blastomeres at the late 2-(F) and 4-cell (G) stage after microinjection of siHspa2 in zygotes. *Hprt* as the housekeeping gene. CT values are used for level analysis. Two-tailed Student's t tests were used for the statistical analysis. Blastomeres are named B1 to B4 (F) or B1 and B2 (G) according to increasing *Hspa2* concentration. **p* <0.05.

Figure S2.

Knockdown of *Hspa2* prevents blastomeres from undergoing an ICM fate. Related to Figure 3.

(A) Representative images of preimplantation embryos in the NC and KD groups from 2-cell to blastocyst. **(B)** Bar plots showing the developmental rates of NC group and KD group at the blastocyst stage.

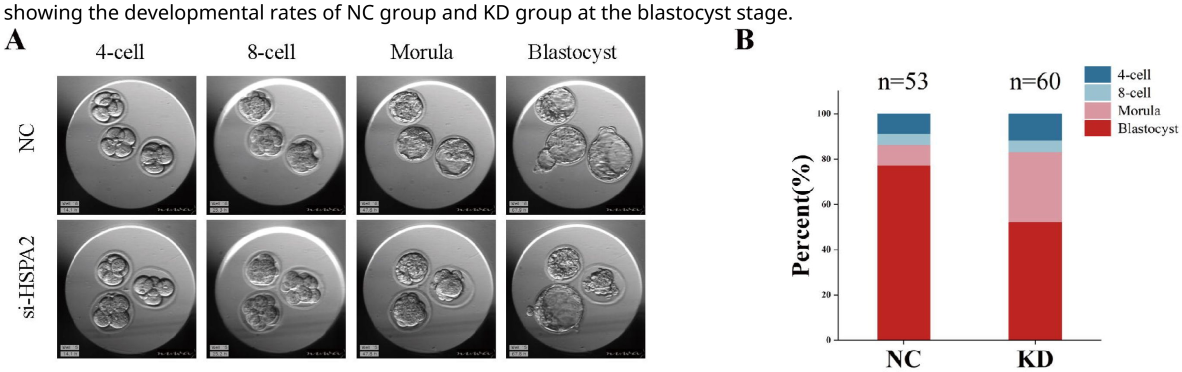
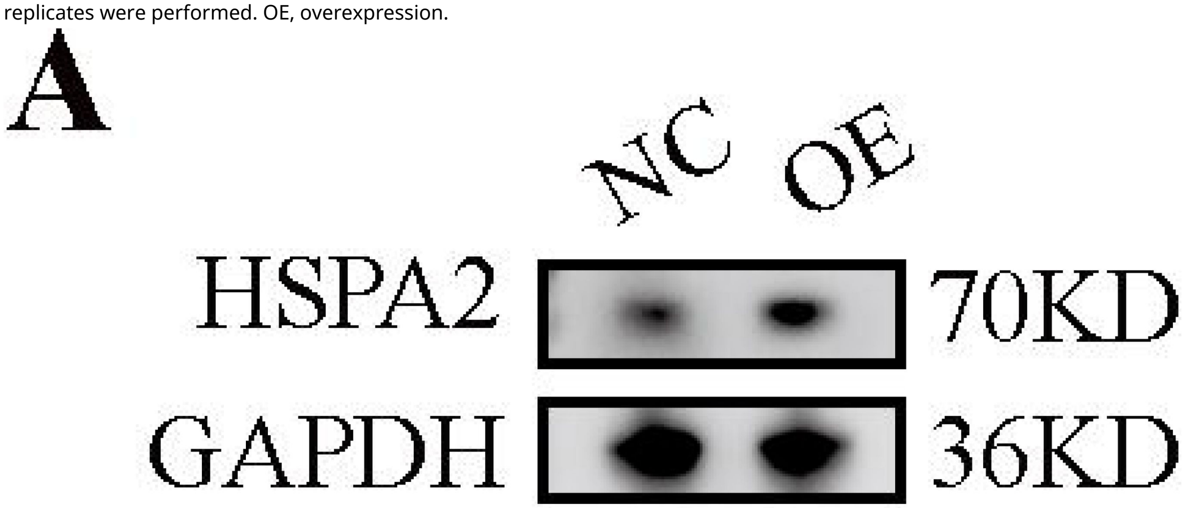


Figure S3.

Overexpression of *Hspa2* not induces blastocyst bias towards ICM fate. Related to Figure 4.

(A) Western blot results confirming the overexpression of HSPA2 protein in the blastocyst stages. Three experimental replicates were performed. OE, overexpression.



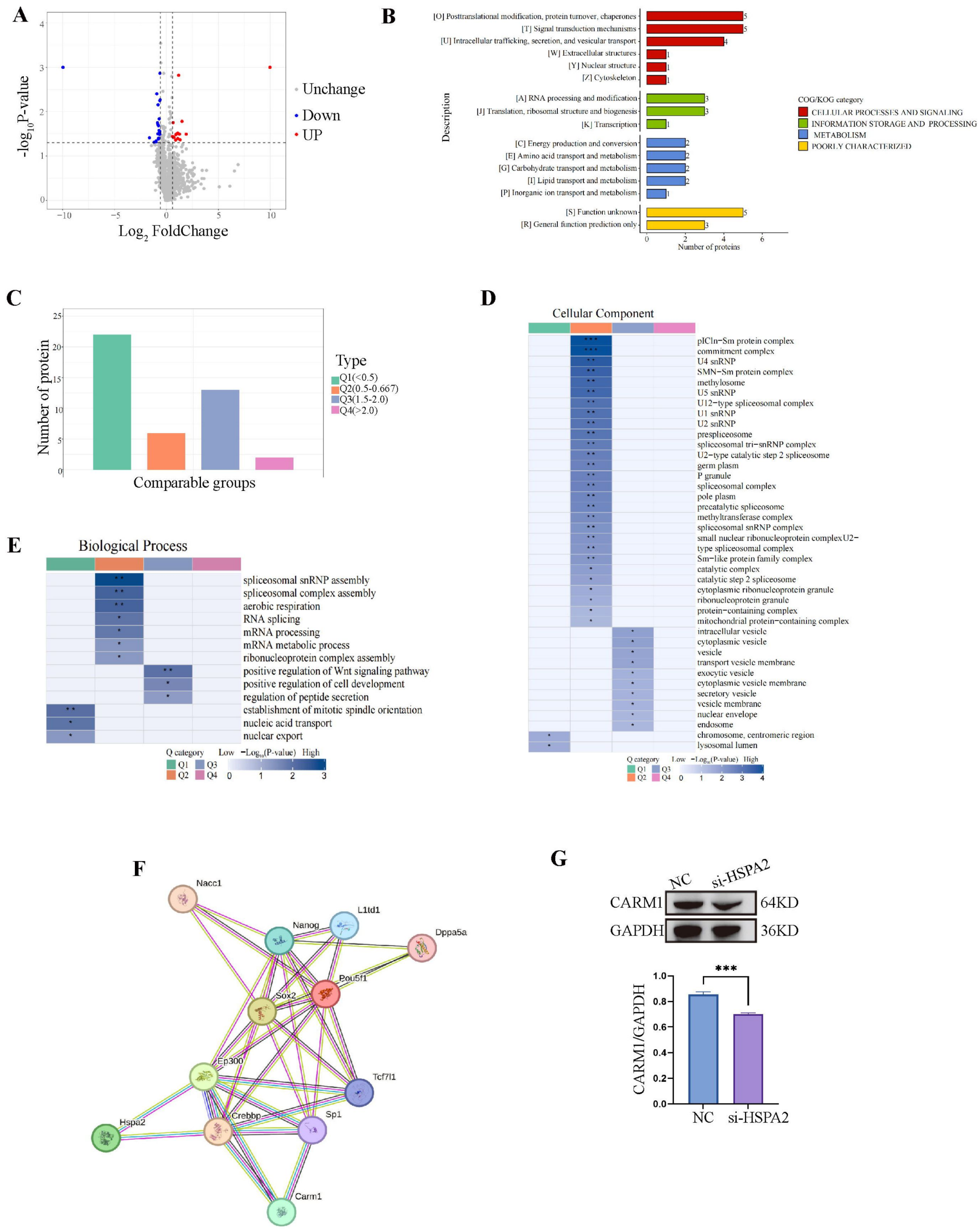


Figure S4.

HSPA2 interacts with CARM1 and alters H3R26me2 levels. Related to Figure 5.

(A) Volcano plot of the DEPs in *Hspa2* siRNA and control groups at blastocyst stage. **(B)** COG/KOG enrichment analysis for DEPs. **(C)** Quantitative classification and numbers of DEPs in each quantified part. The DEPs were divided into four quantified parts according to their differential expression multiples: Q1 (<0.5), Q2 (0.5-0.667), Q3 (1.5-2.0), and Q4 (>2.0). **(D and E)** Clustering analysis including biological process analysis (D) and cellular component (E) of quantitative DEPs in each Q group. **(F)** The Protein-protein interactions (PPIs) network was mapped by STRING database (*https://cn.string-db.org/cgi/network?taskId=bk9UNYyN0p0f&sessionId=bS81YN78aR*). **(G)** Western blot analysis of CARM1 expression in the knockdown and control groups. The band intensity was assessed with ImageJ. Three experimental replicates were performed. ***$p < 0.001$.

Acknowledgements

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Additional information

Compliance with Ethical Standards

The current study was carried out with the approval of the Ethic Committee of Reproductive Medicine of Reproductive Hospital of Shandong University (2021-15), and all experiments performed were abided by relevant regulations and guidelines of the committees.

Author contribution

Jiayin Gao wrote the first draft of the manuscript. Jiawei Wang modified the language. All authors participated in literature research, conception, design, and discussion of the manuscript, and approved its final version.

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Reviewer #1 (Public review):

Summary:

The authors investigate the role of HSPA2 during mouse preimplantation development. Knocking down HSPA2 in zygotes, the authors describe lower chances of developing into blastocysts, which show a reduced number of inner cell mass cells. They find that HSPA2 mRNA and protein levels show some heterogeneity among blastomeres at the 4-cell stage and propose that HSPA2 could contribute to skewing their relative contribution to embryonic lineages. To test this, the authors try to reduce HSPA2 expression in one of the 2-cell stage blastomere and propose that it biases their contribution to towards extra-embryonic lineages. To explain this, the authors propose that HSPA2 would interact with CARM1, which controls chromatin accessibility around genes regulating differentiation into embryonic lineage.

Strengths:

(1) The study offers simple and straightforward experiments with large sample sizes.

(2) Unlike most studies in the field, this research often relies on both mRNA and protein levels to analyse gene expression and differentiation.

Weaknesses:

(1) Image and statistical analyses are not well described.

(2) The functionality of the overexpression construct is not fully validated.

(3) Tracking of KD cells in embryos injected at the 2-cell stage with GFP is unclear.

(4) A key rationale of the study relies on measuring small differences in the levels of mRNA and proteins using semi-quantitative methods to compare blastomeres. As such, it is not possible to know whether those subtle differences are biologically meaningful. For example, the lowest HSPA2 level of the embryo with the highest level is much higher than the top cell from the embryo with the lowest level. What does this level mean then? Does this mean that some blastomeres grafted from strong embryos would systematically outcompete all other blastomeres from weaker embryos? That would be very surprising. I think the authors should be more careful and consider the lack of quantitative power of their approach before reaching firm conclusions. Although to be fair, the authors only follow a long trend of studies with the same intrinsic flaw of this approach.

(5) Some of the analyses on immunostaining do not take into account that this technique only allows for semi-quantitative measurements and comparisons.
a) Some of the microscopy images are shown with an incorrect look-up table.
b) Some of the schematics are incorrect and misleading.

https://doi.org/10.7554/eLife.100730.2.sa2

Reviewer #2 (Public review):

Summary:

In this study, Gao et al. use RNA-seq to identify Hspa2 as one of the earliest transcripts heterogeneously distributed between blastomeres. Functional studies are performed using siRNA knockdown showing Hspa2 may bias cells toward the ICM lineage via interaction with the known methyltransferase CARM1.

Strengths:

This study tackles an important question regarding the origins of the first cell fate decision in the preimplantation embryo. It provides novelty in its identification of Hspa2 as a heterogeneous transcript in the early embryo and proposes a plausible mechanism showing interactions with Carm1. Multiple approaches are used to validate their functional studies (FISH, WB, development rates, proteomics). Given only 4 other transcripts/RNA have been identified at or before the 4-cell stage (LincGET, CARM1, PRDM14, HMGA1), this would be an important addition to our understanding of how TE vs ICM fate is established.

Weaknesses:

The RNA-seq results leading the authors to focus on Hspa2 are not included in the manuscript. This dataset would serve as an important resource but is neither included nor discussed. Nor is it mentioned whether Hspa2 was identified in prior RNA-seq embryos studies (for example Deng Science 2014).

Furthermore, the authors show that Hspa2 knockdown at the 1-cell stage lowers total Carm1 levels at the 4-cell stage. However, it is unclear how total abundance within the embryo alters lineage specification within blastomeres. The authors go on to propose a plausible mechanism involving Hspa2 and Carm1 interaction, but do not discuss how expression levels may be involved.

https://doi.org/10.7554/eLife.100730.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The authors investigate the role of HSPA2 during mouse preimplantation development. Knocking down HSPA2 in zygotes, the authors describe lower chances of developing into blastocysts, which show a reduced number of inner cell mass cells. They find that HSPA2 mRNA and protein levels show some heterogeneity among blastomeres at the 4-cell stage and propose that HSPA2 could contribute to skewing their relative contribution to embryonic lineages. To test this, the authors try to reduce HSPA2 expression in one of the 2-cell stage blastomere and propose that it biases their contribution to towards extra-embryonic lineages. To explain this, the authors propose that HSPA2 would interact with CARM1, which controls chromatin accessibility around genes regulating differentiation into embryonic lineage.*
>
> *Strengths:*

> *(1) The study offers simple and straightforward experiments with large sample sizes.*

Thanks for your kind recognition.

> *(2) Unlike most studies in the field, this research often relies on both mRNA and protein levels to analyses gene expression and differentiation.*

Thanks for your kind recognition.

> *Weaknesses:*
>
> *(1) Image and statistical analyses are not well described.*

Thanks for your advisable comment. We redescribe the image and statistical analyses in our revised version (line 255-257).

> *(2) The functionality of the overexpression construct is not validated.*

Thanks for your kind suggestion. We validate the functionality of the overexpression construct in our revised version (Figure S3).

> *(3) Tracking of KD cells in embryos injected at the 2-cell stage with GFP is unclear.*

Thanks for your kind suggestion. We randomly co-injected green fluorescent protein (Gfp) mRNA as a linage tracer with either Hspa2-siRNA or NC-FAM into one of the 2 -cell, and then monitored embryo development to the blastocyst stage (line 342-344).

> *(4) A key rationale of the study relies on measuring small differences in the levels of mRNA and proteins using semi-quantitative methods to compare blastomeres. As such, it is not possible to know whether those subtle differences are biologically meaningful. For example, the lowest HSPA2 level of the embryo with the highest level is much higher than the top cell from the embryo with the lowest level. What does this level mean then? Does this mean that some blastomeres grafted from strong embryos would systematically outcompete all other blastomeres from weaker embryos? That would be very surprising. I think the authors should be more careful and consider the lack of quantitative power of their approach before reaching firm conclusions. Although to be fair, the authors only follow a long trend of studies with the same intrinsic flaw of this approach.*

Thanks for your advisable comment. Indeed, despite the approach drew on previous research (Zhou Cell 2018), we were clearly aware that this approach can only reflect relative comparisons. This means that the relative difference among the blastomeres from the same embryo were detected and compared. We did not compare the absolute levels of mRNA between different embryos. We also offered simple and straightforward experiments with large sample sizes to confirm this conclusion.

> *(5) Some of the analyses on immunostaining do not take into account that this technique only allows for semi-quantitative measurements and comparisons.*
>
> *a) Some of the microscopy images are shown with an incorrect look-up table.*
>
> *b) Some of the schematics are incorrect and misleading.*

Thanks for your advisable comment. We revised microscopy images and schematics in our revised version.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *In this study, Gao et al. use RNA-seq to identify Hspa2 as one of the earliest transcripts heterogeneously distributed between blastomeres. Functional studies are performed using siRNA knockdown showing Hspa2 may bias cells toward the ICM lineage via interaction with the known methyltransferase CARM1.*
>
> *Strengths:*
>
> *This study tackles an important question regarding the origins of the first cell fate decision in the preimplantation embryo. It provides novelty in its identification of Hspa2 as a heterogeneous transcript in the early embryo and proposes a plausible mechanism showing interactions with Carm1. Multiple approaches are used to validate their functional studies (FISH, WB, development rates, proteomics). Given only 4 other transcripts/RNA have been identified at or before the 4-cell stage (LincGET, CARM1, PRDM14, HMGA1), this would be an important addition to our understanding of how TE vs ICM fate is established.*

Thanks for your kind recognition.

> *The RNA-seq results leading the authors to focus on Hspa2 are not included in the manuscript. This dataset would serve as an important resource but is neither included nor discussed. Nor is it mentioned whether Hspa2 was identified in prior RNA-seq embryos studies (for example Deng Science 2014).*

Thanks for your advisable comment. To identify genes that show a significantly high variability across blastomeres in the same embryo, we regressed out the embryo effect by established a new method, which will be published and uploaded to the database in the future. Thus, the RNA-seq results leading the we focus on Hspa2 are not included in the manuscript.

> *In addition, the functional studies are centered on Hspa2 knockdown at the zygote (1-cell) stage, which would largely target maternal transcript. Given the proposed mechanism relies on Hspa2 heterogeneity post-ZGA (late 2-cell stage), the knockdown studies don't necessarily test this and thus don't provide direct support to the authors' conclusions. The relevance of the study would be improved if the authors could show that zygotic knockdown leads to symmetric Hspa2 levels at the late 2-cell and/or 4-cell stage. It may be possible that zygotic knockdown leads to lower global Hspa2 levels, but that asymmetry is still generated at the 4-cell stage.*

Thanks for your advisable comment. We showed that the *Hspa2* levels at the late 2-cell and 4cell stage after zygotic knockdown in our revised version (Figure S1 G-H, line 450-452).

> *Furthermore, the authors show that Hspa2 knockdown at the 1-cell stage lowers total Carm1 levels at the 4-cell stage. However, it is unclear how total abundance within the embryo alters lineage specification within blastomeres. The authors go on to propose a plausible mechanism involving Hspa2 and Carm1 interaction, but do not discuss how expression levels may be involved.*

Thanks for your advisable comment. Previous research suggests that heterogeneous activity of the methyltransferase CARM1 results in differential methylation of histone H3R26 to modulate establishment of lineage specification (Zernicka-Goetz Cell 2018). Thus, we didn't discuss the total abundance within the embryo alters lineage specification.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *(1) Major issue with analyses:*
>
> *Image analysis needs to be much better explained than simply saying that ImageJ was used. Where are cells measured (at their equatorial plane? What is the size of the ROI?)? Ideally, the ROI and/or raw measurements should be provided.*

Thanks for your advisable comment. We redescribe the Image analysis in our revised version (line 187-194).

> *What are the objective criteria determining whether a cell is counted as GFP positive, CDX2 positive, or OCT4 positive? This is very unclear and key to the interpretation of many experiments.*

Thanks for your advisable comment. We think that the cell containing fluorescence signals above background noise were counted positive.

> *Statistical analyses mention ANOVA in the methods but the student's t-test in the figure legend. Which is which? Most data are heavily normalized, which would unlikely fit the description for Student's t-test analyses.*

Thanks for your advisable comment. We redescribe the statistical analyses in our materials and methods (line 253-260).

> *Figure 5H describes a relative fluorescence intensity with control at 1. The legend describes a normalization to "DNA" (I guess the authors meant DAPI), which is unlikely to give 1. This suggests that additional normalization was done and is not described. Is that the case? Also, since the authors propose that HSPA2 would control Histone modification and chromatin packing, I do not think that using DAPI is an appropriate way of normalizing the fluorescence signal.*

Thanks for your advisable comment. We replaced DNA with DAPI in our revised version. Based on previous studies, we adopted DAPI as a normalized fluorescence signal (Zhou Cell 2018, Zernicka-Goetz Cell 2018).

> *Figure 1E shows data normalized to the lowest level while Figure 1H is normalized to the highest level. A consistent representation would be welcome.*

Thanks for your advisable comment. We revised the Figure 1H in our revised version.

> *Is Figure 1C showing a t-test between correlations?*

Yes, Figure 1C shows the t-test between correlation.

> *(2) Major issue with the interpretation of semi-quantitative methods and measurements:*
>
> *qPCR, WB, immunostaining are all semi-quantitative methods that require some kind of normalization due to non-linear bias in the way the molecules are picked up. Such normalization makes it difficult to know whether a detectable difference is meaningful*

> *biologically speaking i.e. if a difference of 1 CT between blastomeres can be detected after qPCR, is it meaningful? If that were the case, then embryos with lower CT than others (Figure 1D) would not be able to develop into blastocyst, like siRNA injected embryos, or grafting a blastomere with a high CT onto an embryo with low CT would lead to the systematic differentiation of these strong blastomeres into ICM.*

Thanks for your advisable comment. The CT values represent the relative mRNA levels of *Hspa2* between blastomeres, and the higher CT value represents the lower expression of *Hspa2* at mRNA level. Figure 1D shows the *Hspa2* mRNA levels between blastomeres. The blastomere with lowlevel expression of the *Hspa2* mRNA is not bias an ICM fates.

> *The same goes for fluorescence analyses (Figure 1F). Can the authors also provide the measurements for DAPI as they did for HSPA2? I am sure that with enough measurements, DAPI is variable enough to give a statistical difference among blastomeres with questionable biological meaning.*
>
> *I think the reasoning used here (unfortunately following the reasoning that has been used in a series of studies by other groups) of ranking blastomeres after semi-quantitative measurement is fundamentally flawed.*

Thanks for your advisable comment. The DAPI was determined by the maximal area using a custom Python script. Based on previous studies, we adopted DAPI as a normalized fluorescence signal (Zhou Cell 2018). This approach is to normalize embryo-to-embryo variance from the technical reason.

> *(3) Major issue with overexpression experiment:*
>
> *While the siRNA experiment is partially validated by qPCR and WB measurements of HSPA2 after KD, the overexpression experiment is not. Do the authors have any evidence that the construct they use is produced into protein and functional? Can the authors check by WB? Can the authors rescue the siRNA with their overexpression?*

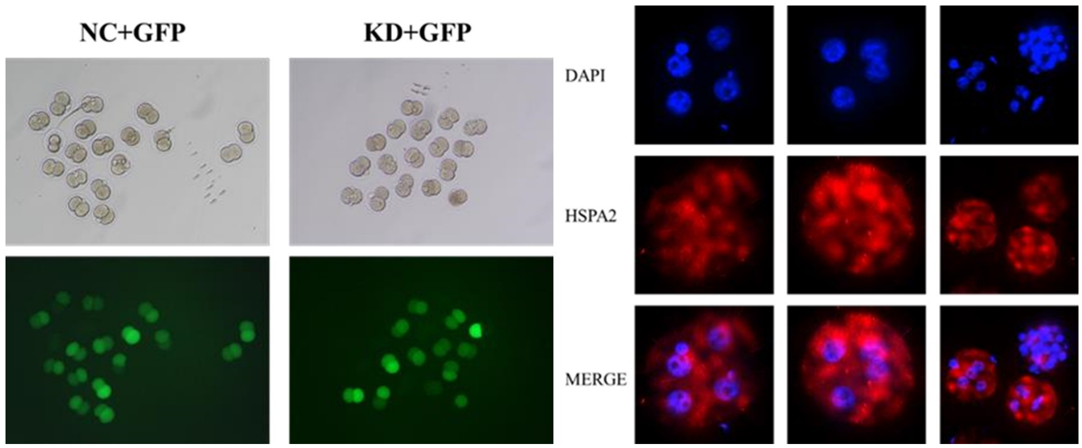
Thanks for your advisable comment. We verified the overexpression experiment by WB in in our revised version (Figure S3, line 360-361). Considering that siRNA degrades mRNA and prevents the mRNA translation process, we did not co-inject the siRNA with their overexpression.

The lack of effect of HSPA2 overexpression on blastocyst formation is difficult to reconcile with the interpretation from the authors that levels of HSPA2 bias lineages.

> *Have the authors tried lower concentrations? Have the authors tried FISH on their half-injected 2cell embryos? Of course, if the antibody against HSPA2 would work with immunostaining, that would be ideal.*

Thanks for your advisable comment. We chose the concentrations for our study based on previous research (Zernicka-Goetz Cell 2016). To verified Hspa2 was successfully inject into one blastomere at the 2-cell stage, we observed green fluorescence after co-injected GFP mRNA with either siRNA or NC-FAM into one blastomere of the two-cell embryos. Thus, we didn't try FISH on half-injected 2-cell embryos. We tried to perform immunostaining experiments with various HSPA2 antibodies (Proteintech: 12797-1-AP, Abcam: ab108416) and no good results were achieved.

Author response image 1.



> *(4) Major issue with tracking of injected cells:*
>
> *It is unclear what counts as a GFP-positive cell. In Figure 3D, most cells appear to have the same level of GFP.*

Thanks for your advisable comment. The cell containing green fluorescence signals above background noise were counted GFP-positive in Figure 3D. Most cells seem to have the same level of GFP because they are daughter cells of the blastomeres injected with GFP.

> *In the images of GFP-expressing cells used to track the control of KD cells shown in Figure 3A, it seems that the control embryos have mostly GFP cells in the ICM. Is that the case, or just a bad example?*

Thanks for your advisable comment. The green fluorescent signals in Figure 3A represented OCT4 protein, an ICM marker.

> *Can the authors do FISH against HSPA2 and visualize their GFP cells to validate the heterogeneous expression in situ?*

Thanks for your advisable comment. We have verified the heterogeneous expression of HSPA2 in Figure1.

> *(5) Issue with fluorescent images:*
>
> *Many images are shown with inappropriate look-up tables with saturated DAPI, OCT4, CDX2, and FISH. This raises the doubt that analyses were made on saturated images, which would be incorrect.*

The LUT of Figure 5H should be adjusted similarly between the control and siRNA.

Thanks for your advisable comment. We revised some images which showed inappropriate lookup tables in our revised version. The LUT of Figure 5H had been adjusted between the control and siRNA.

> *(6) Issue with schematics:*

> *Schematics of blastomere isolation grown into blastocyst-like structures are misleading since the final blastocyst-like structure should not have a zona pellucida and should have fewer cells than regular blastocysts.*

Thanks for your advisable comment. We revised schematics of blastomere grown into morula in our revised version (Figure 1A and Figure S1A).

> *The summary schematics in the final figure should not state HSPA2 -/- since experiments in the study did not use KO but KD.*

Thanks for your advisable comment. We revised the summary schematics in our revised version.

> *The blastocysts are the same sizes as the cleavage stage or morula embryos which implies that cells lose volume to the lumen, which is not the case.*

Thanks for your advisable comment. We revised the schematics in our revised version.

> *(7) Issue with data presentation:*
>
> *In the tables within the figures, the number of decimals given should be the same for the mean and SE (one decimal should be more than enough).*

Thanks for your advisable comment. We revised the figure 2H in our revised version.

> *The comparison of cell number and distribution within embryos (e.g. Figure 2B) would be best represented by a correlation analysis of TE vs ICM cells.*

Thanks for your advisable comment. We add the figure of a correlation analysis of TE vs ICM cells in our revised version (Figure 3B).

> *The docking simulations are described in the main text as "experiments".*

Thanks for your advisable comment. We redescribed the docking simulations in our revised version.

> *(8) Issue with data interpretation:*
>
> *The reduced number of ICM cells is interpreted as a slowed-down cell cycle. This could also be explained by failed cytokinesis and the generation of binucleated or polyploid cells. Have the authors checked for that? For example, by looking at their DAPI staining.*

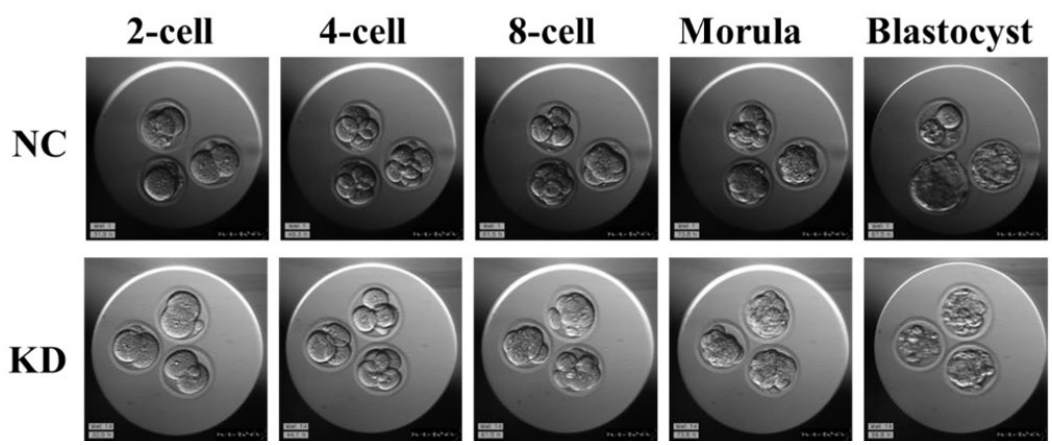
Thanks for your advisable comment. Our RNA-seq results revealed that the differentially expressed genes (DEGs) at blastocyst stage with HSPA2 knocking down are closely related to negative regulation of cell cycle, G1/S transition of mitotic cell cycle, mitotic cell cycle phase transition and regulation of mitotic cell cycle phase transition. Additionally, the previous study demonstrated that knockdown of HSPA2 reduced cell proliferation and led to G1/S phase cell cycle arrest (Hu Ann Transl Med 2019). Additionally, the lower cell number in ICM may also associated with failed cytokinesis and the generation of binucleated or polyploid cells. Thus, we guessed that HSPA2 has a role in ICM lineage establishment, although half of the ICM cells were able to survive with HSPA2 deficiency (line 463-472).

> *It is unclear to me why reduced ICM should lead to fewer blastocysts. Blastocysts should be able to form as long as their TE is fine. In Figure 2G, embryos seem to be cultured in*

> *close proximity, which is fine if they are healthy but not if some of the embryos start dying and releasing toxic compounds (e.g. ROS). Have the authors tried removing the dying KD embryos to see if the development of the remaining embryos would improve?*

Thanks for your advisable comment. We think HSPA2 may affect blastocyst development by affecting other signaling pathways. And, the GO enriched terms was closely related to blastocyst development (Figure 2E). There was no significant difference in morula formation rate between *Hspa2*-KD group and NC group, thus the assumption that the toxic compounds released by some of the embryos that lead to downregulation of blastocyst rate may not be correct. Indeed, the rate of blastocyst formation in *Hspa2*-KD embryos was reduced significantly lower when few embryos was cultured separately. In addition, we discussed the possibility that the lower cell number in ICM may also associated with failed cytokinesis and the generation of binucleated or polyploid cells.

Author response image 2.



> ***Reviewer #2 (Recommendations for the authors):***
>
> *One of the significant findings in the paper is the discovery portion where Hspa2 is identified as a heterogeneous transcript. To improve the logic and impact of the manuscript, it may benefit from reorganizing some of the figures and text. For example:*
>
> *(1) The paragraph in the introduction (Lines 56-68) should be moved to the discussion as the Hspa2 reveal should be in section 3.1, not prior to the RNA-seq results presented in Figure 1.*

Thanks for your advisable comment. We think it is more logical that HSPA2 needs to be introduced in the introduction.

> *(2) Add text at the beginning of Section 3.1 to describe the rationale and results for the RNAseq. It would help the readers if the authors clearly stated why they chose the 4-cell stage.*

Thanks for your advisable comment. We explain why we chose the 4-cell stage in our revised version (line 272-273).

> *(3) As this is the first time Hspa2 is identified, consider moving Figure S1C to the main figure to show expression throughout development.*

Thanks for your advisable comment. We moved Figure S1C to the main figure in our revised version (line 286-291).

> *(4) Figure 1C: the correlation between Hspa2 and ICM markers would be strengthened if additional transcripts were used (Oct4, Sox2, Sox21). The graph in 1C would also be more informative if represented as a scatter plot with correlation coefficients (Nanog log2TPM vs Hspa2 log2TPM), rather than bar graphs.*

Thanks for your advisable comment. We chose Nanog as the correlation between Hspa2 and Nanog, a ICM markers, was showing the strongest correlation in result. And, the figure 1C shows the stronger positive correlation between *Nanog* and *Hspa2* in gene expression than random gene pairs (n=100, n means the number of random gene pairs). Thus, the figure 1C with bar graphs is easier to understand.

> *(5) Figure 1D: how were individual blastomeres grouped into B1-4? Individually run and then pooled based on relative expression?*

Thanks for your advisable comment. Blastomeres are named B1 to B4 according to increasing Hspa2 concentration in figure 1E.

> *(6) Figures 1F, 1I, 5H: the DAPI channel appears to be saturated, but is used to normalize fluorescence intensity and may incorrectly account for light scattering within the embryo. Please clarify by adding more details regarding image analysis. Were partial stacks through the nucleus used for analysis, or max projections? Graph axes should be "relative fluorescence intensity."*

Thanks for your advisable comment. We added the details of fluorescence images analysis. The graph axes had revised in our revised version.

> *(7) Line 278: the results in Figure S1C would benefit from more text regarding expression patterns throughout development. The maternal transcript appears to have a sharp downregulation by the early 2-cell stage, and is then upregulated coinciding with ZGA.*

Thanks for your advisable comment. We added more describe of the Figure in main text (LINE 285-290).

> *(8) For the analyses in Figure 2 I-J and 2K-L, were arrested embryos excluded from analysis? This is an important detail as including arrested embryos would significantly bias the RNA-seq results.*

Thanks for your advisable comment. The arrested embryos were excluded in Figure 2 I-J and 2K-L.

> *(9) Figures 2G-H would be aided by converting the table in 2H to a bar graph and adding development rates for all stages (2-, 4-, 8-, morula, and blast). This would also show when an arrest occurs.*

Thanks for your advisable comment. We converted the table in 2H to a bar graph.

> *(10) Blast rates are represented with too many significant digits (Figures 2H, 4B). They should only be reported to the closest ones given the unit of measure (number of blasts divided by number of zygotes). For instance, a blast rate of 81.63 {plus minus} 2.000*

> *reflects excessive precision that is not measured in the data, it should rather read 82 {plus minus} 2%. This is also true for % cells (Figures 3E, 4H).*

Thanks for your advisable comment. Values were rounded down to the one decimal place (rounded down).

> *(11) The clarity and impact of Figure 3A and 3D would benefit from 2D slices through the ICM.*

Thanks for your advisable comment. In order to get more comprehensive understanding of the 3D structure of blastocyst of Figure 3A and 3D, we did not choose 2D slices.

> *(12) To improve clarity and logic, separate the 1-cell and 2-cell knockdown experiments in the text and figures:*
>
> *a) 1-cell knockdown with RNA-seq results (Fig 2A-F).*
>
> *b) 1-cell knockdown showing less ICM/pluripotency markers in (combine Figures 2G-M and Figures 3A-B; "new Fig 3").*
>
> *c) 2-cell knockdown tracing lineage (Figures 2D-E; "new Fig 4").*
>
> *The new Figures 3 and 4 should mirror one another (i.e. for each knockdown experiment, development rates and cell counts should be included). For the 2-cell knockdown (Figures 2 D-E), what were the developmental rates (8-cell, morula, blast)?*

Thanks for your advisable comment. However, in order to the overall logical of the article, we do not separate the 1-cell and 2-cell knockdown experiments in the text and figures. And, we added the developmental rates (8-cell, morula, blast) of 2-cell knockdown group in our revised version (Figure S2).

> *For the overexpression experiment (Figure 4), why were injections performed at the zygote stage versus the 2-cell stage? Given the significant downregulation of maternal transcript demonstrated in Figure S1C, it seems plausible that the injected RNA was also downregulated.*

Thanks for your advisable comment. For the overexpression experiment, we first chose to inject *Hspa2* mRNA at the zygote stage and found that the overexpression of Hspa2 does not induce blastomere cells to bias an ICM fate. The qRT-PCR results indicated that the expression level of *Hspa2* in overexpression group was significantly increased compared with normal group at 4cell and blastocyst stage (Figure 4C, 4D). In addition, there is no guarantee that an equal amount of *Hspa2* mRNA be injected into each blastomere in 2-cell stage. Thus, we did not microinject *Hspa2* mRNA into the 2-cell stage.

> *The 3.5 subheading overstates the results as the Hspa2-Carm1 interaction is not linked to lineage segregation. For example, a more specific subtitle might be, "Hspa2 interacts with Carm1 and alters H3R26me2 levels."*

Thanks for your advisable comment. We revised the subtitle in our revised version (line 376).

> *Figures 5B-C and 5D-E. The qRT-PCR and WB analysis of knockdown blasts shows a correlation between Hspa2 downregulation and Carm1 downregulation. However, if the proposed mechanism is Hspa2 binding to Carm1 to mediate downstream methylation, why would it be expected to alter transcript levels at the 4-cell or blast stage? Please add further details and discussion in the results and discussion sections.*

Thanks for your advisable comment. The reason we chose to work at the 4-cell stage is because previous studies on CARM1 have focused on the 4-cell stage (Zernicka-Goetz Cell 2018,2016).

> *In the discussion, the statement in Lines 430-431 is an overinterpretation: "the heterogeneity of HSPA2... acts as an upstream factor to drive [the] first cell-fate decision." The knockdown experiments don't alter heterogeneity per se, but total abundance. Furthermore, the results do not show that heterogeneity drives heterogeneity in H3R26me2 patterns, for example.*

Thanks for your advisable comment. We redescribe the relevant statement in the discussion.

More needs to be said regarding the ICM cells that persisted in the 1-cell KD experiment (Fig 3B). Lines 449-450 point out this result, but do not propose any plausible explanations. For instance, ICM cells may still form due to the incomplete knockdown achieved or the possibility that redundant pathways exist.

> *Thanks for your advisable comment. We redescribe the relevant statement in our revised version (line 468-473).*

The 5th paragraph of the discussion seems incomplete. The authors point out a possible link between Hspa2 and Hippo and Wnt signaling pathways, but need to expand their discussion on how this may act as an additional mechanism incorporating Hspa2 with lineage segregation.

> *Thanks for your advisable comment. We redescribe the 5th paragraph of the discussion (line 483-494).*

Statistics: all comparisons with greater than 2 groups should be performed with a one-way ANOVA and multiple comparisons, rather than Student's t-test (Figures 1B, 1D, 1E, 1F).

All figure legends lack statistical test details.

> *Thanks for your advisable comment. All figure legends added statistical test details in statistical analysis.*
>
> *Minor comments:*
>
> *In all graphs, individual blastomere expression levels should be represented as boxwhisker/bar/scatter/violin plots since the comparison is groups rather than time points (i.e. symbols should not be connected with a line in Figures 1B, 1D, 1F-G, 1I, S1D, S1F).*

Thanks for your advisable comment. Each colored line represents a single cell, and the dots of the same color represent the blastomere of the same cell. Thus, we use a line representation individual blastomere.

> *For all fluorescent images, having two representative images may be confusing for the reader. Figures may be improved by just including one representative image for each stage/treatment (Figures 1F, 1I, S1F, 3A, 3D, 4E, 4G).*

Thanks for your advisable comment. The figures just including one representative image for each stage in our revised version. In addition, two representative images from each group were shown for each treatment (Figures 3A, 3D, 4E, 4G).

> *The manuscript would be improved with thorough grammar and typo editing.*
>
> *For example:*
>
> *(1) Lines 18, 73, the wording is confusing, consider: "knockdown of Hspa2 in one of the two-cell blastomeres biased its progeny towards the trophectoderm lineage.".*
>
> *(2) Line 23, overstatement. Consider: "we demonstrated that HSPA2 levels correlate with ICMassociated genes and that it interacts with the CARM1.".*
>
> *(3) Line 25 confusing wording, "via the execution of commitment and differentiation phases.".*
>
> *(4) Line 37, replace "that" with "of;" replace "cell-fate decisions" with "cell-fate decision".*
>
> *(5) Line 40: needs space before (CARM1).*
>
> *(6) Line 43: the wording is confusing, consider "can result in higher expression levels of".*
>
> *(7) Line 45: wording, consider "Recent [studies have] further suggested".*
>
> *(8) Line 70: plurality, consider "analyzed gene expression pattern".*
>
> *(9) Line 73 typo: "prevents its".*
>
> *(10) Line 76-77 wording, consider "Hspa2 expression patterns can bias cell fate in the mouse embryo".*
>
> *(11) Line 276: remove "in whole embryos," since MII eggs are not embryos.*
>
> *(12) Line 617 "There" should be "Three".*
>
> *(13) Axis label in Fig 3b "Totle" should be "Total".*
>
> *(14) Lines 417, 419 missing spaces.*
>
> *(15) Line 448 missing word, "interfering [with] the cell cycle".*
>
> *(16) Line 462 incorrect word, "[a]polar cells being specified as ICM".*
>
> *(17) Line 469 incorrect plural, "cell differentiation".*

Thanks for your advisable comment. We revised the whole manuscript carefully according to the reviewers' suggestions.

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